Efficient estimation and the cost of complete-case coarsening under monotone sequential MAR

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Abstract

Complete-case coarsening discards observed confounder values from partially complete records. We study its consequences for average treatment effect estimation with two ordered, partially observed confounders under monotone sequential missing at random. We specialize the standard coarsening-at-random transformation to the causal influence function, establish the canonical gradient, and give an exact drift identity for a cross-fitted estimator with sequential multiple robustness. In the submodel where both sequential and complete-case missing-at-random assumptions hold, we express the efficiency loss from coarsening as a nonnegative expectation involving two iterated projections. The gain is strict when the intermediate confounder supplies residual information where second-stage missingness occurs. Oracle simulations and deterministic quadrature illustrate this efficiency comparison. When second-stage response depends on the intermediate confounder, the comparison instead concerns identification: coarsening can introduce persistent bias. Estimated-nuisance simulations include a bounded-propensity design and a Gaussian stress design. The latter exhibits substantial interval undercoverage and can reverse the finite-sample precision ordering, qualifying the practical interpretation of the efficiency bound.

Résumé

Le regroupement des profils de données manquantes selon le seul indicateur de cas complet élimine des valeurs observées de variables de confusion. Nous étudions ses conséquences pour l'estimation de l'effet moyen d'un traitement lorsque deux variables de confusion ordonnées sont partiellement observées sous une hypothèse de données manquantes au hasard de manière séquentielle et monotone. Nous adaptons la transformation classique de données regroupées au hasard à la fonction d'influence causale, établissons le gradient canonique et présentons une identité exacte du biais pour un estimateur à ajustement croisé possédant une propriété de robustesse multiple séquentielle. Dans le sous-modèle où les hypothèses séquentielle et fondée sur les cas complets sont toutes deux valides, la perte d'efficacité s'exprime comme une espérance non négative faisant intervenir deux projections itérées. Des simulations et une intégration numérique déterministe illustrent cette comparaison. Lorsque l'observation au second stade dépend de la variable intermédiaire, le regroupement peut introduire un biais persistant. Des simulations avec fonctions auxiliaires estimées distinguent un modèle à probabilités de traitement uniformément bornées d'un modèle gaussien présentant une sous-couverture importante des intervalles.

Key words and phrases: Average treatment effect; efficient influence function; missing confounders; monotone missingness; multiple robustness; sequential MAR

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1 Introduction

Observational studies frequently combine two statistical difficulties: treatment is confounded, and one or more baseline confounders are missing. Complete-case analysis is generally inefficient and can be biased. Multiple imputation, inverse-probability weighting, augmented estimators, generalized raking, and targeted learning address parts of this problem, but their validity depends on how the causal and missingness assumptions fit the observed-data structure (Bang and Robins, 2005; Williamson et al., 2012; Seaman and White, 2014; Levis et al., 2025; Benz et al., 2025; Williamson et al., 2026; Wen and McGee, 2026).

Levis et al. (2025) developed robust and efficient ATE estimators under complete-case missing at random (CCMAR). If L_c denotes fully observed confounders, L_p the vector of partially observed confounders, and S the indicator that every component of L_p is observed, their assumption is

$$S \perp\!\!\!\perp L_p \mid (L_c, A, Y).$$

The reduction to S avoids modeling a possibly nonmonotone collection of response patterns. It also means that subjects with only part of L_p observed do not directly contribute their observed confounder values to the final estimating function. Levis and colleagues identify the use of such partial records under alternative assumptions as an important direction for future research.

Benz et al. (2025) provide the applied counterpart to that development, comparing the robust estimators of Levis et al. (2025) against ad hoc alternatives—sequential imputation followed by outcome regression, or by inverse-probability weighting—across a range of missing-confounder designs. They report that no single estimator is uniformly best and that the simpler alternatives are adequate in several settings. Their study motivates a practical distinction: an efficiency ordering for influence functions need not translate into a finite-sample precision ordering when nuisance functions are estimated. We study that distinction in Appendix D.

This paper studies the transparent special case in which the components of L_p have a scientifically meaningful order and are observed monotonically. We write

$$L_c = W, \qquad L_p = (L_1, L_2),$$

so the notation L_1, L_2 refines—rather than replaces—the earlier L_c, L_p notation. The possible patterns are: neither partially observed confounder is available; L_1 alone is available; or both are available. Sequential MAR permits the probability of observing L_2 to depend on the already observed L_1. This model is useful when, for example, an inexpensive screening variable precedes a more burdensome laboratory measure, or when data acquisition follows an ordered protocol.

Sequential inverse-probability augmentation and efficient estimation with monotone missingness are established ideas (Robins et al., 1994; Gill et al., 1997; Tsiatis, 2006; Barnwell and Chaudhuri, 2025). Accordingly, our contribution is not the generic sequential augmentation formula. It is the following causal and comparative analysis:

1. We specialize the monotone sequential construction to the full-data influence function for a point-exposure ATE with missing confounders, allowing response probabilities to depend on the fully observed treatment and outcome.
2. We give a self-contained observed-likelihood and tangent-space proof of the efficient influence function, followed by an exact remainder identity and a cross-fitted estimator.
3. On the submodel where sequential MAR and CCMAR are simultaneously valid, we derive the exact nonnegative efficiency loss from collapsing the intermediate pattern.
4. We distinguish this efficiency comparison from the case in which second-stage response depends on L_1. There, CCMAR generally fails, so the comparison concerns identification, bias, and coverage rather than two efficiency bounds for the same model.

The final distinction is practically important. A statement that the pattern-aware method is "more efficient" is justified only under common validity. Outside that submodel, better mean-squared error may result primarily from removing bias under a different identifying assumption. Sequential MAR and unrestricted CCMAR are not generally nested models; within the sequential model, Assumption 3 defines their common-validity submodel.

2 Observed data, causal target, and assumptions

2.1 Notation and observation patterns

For one individual, let

$$Z_0 = (W, A, Y), \qquad Z_1 = (Z_0, L_1), \qquad Z_2 = (Z_1, L_2),$$

where W is a vector of fully observed baseline confounders, $A \in \{0,1\}$ is a fully observed point exposure, Y is a fully observed outcome, and L_1, L_2 are partially observed baseline confounders. Let R_1 indicate that L_1 is observed. Among records with $R_1 = 1$, let C_2 indicate that L_2 is observed, and define $R_2 = R_1 C_2$. Thus

$$(R_1, R_2) \in \{(0,0), (1,0), (1,1)\};$$

the L_2-only pattern $(0,1)$ is excluded. The observed datum is

$$O = (Z_0, R_1, R_1 L_1, R_2, R_2 L_2).$$

Let $V = (W, L_1, L_2)$. The causal estimand is

$$\psi = \mathbb{E}\{Y(1) - Y(0)\}.$$

Assumption 1 (Causal identification)**.** *For $a \in \{0,1\}$: (i) consistency, $Y = Y(A)$; (ii) conditional exchangeability, $Y(a) \perp\!\!\!\perp A \mid V$; and (iii) treatment positivity, $\epsilon_e < e(V) < 1 - \epsilon_e$ almost surely for*

Table 1: Relationship between the original CCMAR notation and the ordered notation used here.

Object	CCMAR notation	Sequential notation
Always observed confounders	L_c	W
Partially observed confounders	L_p	(L_1, L_2)
Complete-case indicator	S	$R_2 = R_1 C_2$
Intermediate partial record	discarded by coarsening	$R_1 = 1, R_2 = 0$

some $\epsilon_e > 0$*, where* $e(v) = \mathbb{P}(A = 1 \mid V = v)$.

Under Assumption 1, writing $\mu_a(v) = \mathbb{E}(Y \mid A = a, V = v)$,

$$\psi = \mathbb{E}\{\mu_1(V) - \mu_0(V)\}. \tag{1}$$

2.2 Sequential missing at random

Assumption 2 (Monotone sequential MAR)**.** *The response process satisfies*

$$R_1 \perp\!\!\!\perp (L_1, L_2) \mid Z_0, \tag{2}$$
$$C_2 \perp\!\!\!\perp L_2 \mid (Z_1, R_1 = 1). \tag{3}$$

Let

$$\pi_1(Z_0) = \mathbb{P}(R_1 = 1 \mid Z_0), \qquad \pi_2(Z_1) = \mathbb{P}(C_2 = 1 \mid R_1 = 1, Z_1),$$

and suppose there is $\epsilon_\pi > 0$ *such that* $\pi_1(Z_0) \geq \epsilon_\pi$ *and* $\pi_2(Z_1) \geq \epsilon_\pi$ *almost surely.*

Treatment and outcome are included in Z_0, so both response probabilities may depend on A and Y. These are MAR/CAR restrictions because the hazards depend only on information observed at the corresponding stage. They are not outcome-independent missing-not-at-random restrictions.

Proposition 1 (Identification of the full-data law)**.** *Under Assumption 2, the full-data density is identified by*

$$p(z_0, l_1, l_2) = p(z_0)\, p(l_1 \mid z_0, R_1 = 1)\, p(l_2 \mid z_1, R_1 = 1, C_2 = 1). \tag{4}$$

Consequently, under Assumption 1, ψ *in Equation (1) is identified from the observed-data law.*

Proof. By Equation (2), $p(l_1, l_2 \mid z_0, R_1 = 1) = p(l_1, l_2 \mid z_0)$, hence $p(l_1 \mid z_0, R_1 = 1) = p(l_1 \mid z_0)$. By Equation (3), among $R_1 = 1$, $p(l_2 \mid z_1, C_2 = 1) = p(l_2 \mid z_1)$. Combining these conditional densities with $p(z_0)$ gives Equation (4). The g-formula is a functional of the identified full-data law. □

Equation (4) is a retrospective factorization because A and Y occur in Z_0. This causes no contradiction with the causal factorization: any identified joint law can also be refactored as $p(v)p(a \mid v)p(y \mid a, v)$, after which Assumption 1 gives the causal interpretation.

3 Efficient influence function under sequential MAR

3.1 Full-data gradient

The efficient influence function for the ATE in the nonparametric full-data model is

$$D_{\mathrm{F}}(Z_2) = G(Z_2) - \psi, \tag{5}$$

where

$$G(Z_2) = \mu_1(V) - \mu_0(V) + \frac{A}{e(V)}\{Y - \mu_1(V)\} - \frac{1-A}{1-e(V)}\{Y - \mu_0(V)\}. \tag{6}$$

Define the iterated projections

$$Q_1(Z_1) = \mathbb{E}\{G(Z_2) \mid Z_1\}, \qquad Q_0(Z_0) = \mathbb{E}\{G(Z_2) \mid Z_0\}. \tag{7}$$

Throughout, $\mathcal{M}$ denotes the observed-data model: the set of laws of O generated by an unrestricted full-data law $p(z_0, l_1, l_2)$ together with unrestricted response probabilities π_1, π_2 satisfying Assumption 2 and its positivity bound. Lemma 1 in Appendix A shows that the tangent space of $\mathcal{M}$ is all of $L_2^0(P_O)$, the mean-zero square-integrable functions of O. The model is therefore nonparametric in the observed-data sense, a pathwise differentiable parameter has exactly one gradient, and that gradient is its efficient influence function.

Theorem 1 (Canonical gradient)**.** *Suppose Assumptions 1 and 2 hold, $D_{\mathrm{F}} \in L_2(P)$, and the response probabilities satisfy the two-sided bounds of Lemma 1. In the nonparametric observed-data model $\mathcal{M}$ induced by monotone sequential MAR, the efficient influence function for ψ is*

$$\boxed{D_{\mathrm{seq}}(O) = Q_0 - \psi + \frac{R_1}{\pi_1}(Q_1 - Q_0) + \frac{R_2}{\pi_1\pi_2}(G - Q_1).} \tag{8}$$

All functions in a term are evaluated only when the variables required for that term are observed.

Proof. We give the observed-likelihood argument. Write

$$p_0(z_0) = p(z_0), \quad p_1(l_1 \mid z_0) = p(l_1 \mid z_0), \quad p_2(l_2 \mid z_1) = p(l_2 \mid z_1).$$

Under Assumption 2, the observed-data likelihood contribution factorizes as

$$\begin{aligned} p_O(o) = p_0(z_0)\{1 - \pi_1(z_0)\}^{1-r_1}\{\pi_1(z_0)p_1(l_1 \mid z_0)\}^{r_1} \\ \times \{1 - \pi_2(z_1)\}^{r_1-r_2}\{\pi_2(z_1)p_2(l_2 \mid z_1)\}^{r_2}. \end{aligned} \tag{9}$$

For regular parametric submodels, the tangent space is the closure of sums of the mutually orthogonal components

$$\begin{aligned}
\mathcal{T}_0 &= \{s_0(Z_0) : \mathbb{E}(s_0) = 0\},\\
\mathcal{T}_1 &= \{R_1 s_1(Z_1) : \mathbb{E}(s_1 \mid Z_0) = 0\},\\
\mathcal{T}_2 &= \{R_2 s_2(Z_2) : \mathbb{E}(s_2 \mid Z_1) = 0\},\\
\mathcal{T}_{\pi_1} &= \{(R_1 - \pi_1) h_1(Z_0)\},\\
\mathcal{T}_{\pi_2} &= \{(R_2 - R_1\pi_2) h_2(Z_1)\}.
\end{aligned}$$

Let $q_1 = \mathbb{E}(D_{\mathrm{F}} \mid Z_1) = Q_1 - \psi$ and $q_0 = \mathbb{E}(D_{\mathrm{F}} \mid Z_0) = Q_0 - \psi$. Candidate Equation (8) can be written

$$D_{\mathrm{seq}} = q_0 + \frac{R_1}{\pi_1}(q_1 - q_0) + \frac{R_2}{\pi_1\pi_2}(D_{\mathrm{F}} - q_1). \tag{10}$$

Lemma 1 shows that these five components are mutually orthogonal and that their closed linear span is the whole of $L_2^0(P_O)$; every tangent direction is therefore a sum of the five types, which is what makes the verification below exhaustive rather than merely suggestive.

The three terms in Equation (10) belong respectively to $\mathcal{T}_0, \mathcal{T}_1$, and $\mathcal{T}_2$, because the required conditional means vanish by iterated expectation; the same argument gives $\mathbb{E}(D_{\mathrm{seq}}) = 0$. Thus D_{seq} belongs to the observed-data tangent space and is orthogonal to the two response-mechanism tangent components.

It remains to verify that D_{seq} represents the pathwise derivative. For $s_0 \in \mathcal{T}_0$,

$$\mathbb{E}(D_{\mathrm{seq}} s_0) = \mathbb{E}(q_0 s_0) = \mathbb{E}(D_{\mathrm{F}} s_0).$$

For a full-law score s_1 satisfying $\mathbb{E}(s_1 \mid Z_0) = 0$, the observed score is $R_1 s_1$, and conditional expectation over R_1, C_2 gives

$$\mathbb{E}(D_{\mathrm{seq}} R_1 s_1) = \mathbb{E}\{(q_1 - q_0)s_1\} = \mathbb{E}(D_{\mathrm{F}} s_1).$$

The last equality uses $\mathbb{E}(s_1 \mid Z_0) = 0$ and $q_1 = \mathbb{E}(D_{\mathrm{F}} \mid Z_1)$. Similarly, for s_2 satisfying $\mathbb{E}(s_2 \mid Z_1) = 0$, the observed score is $R_2 s_2$ and

$$\mathbb{E}(D_{\mathrm{seq}} R_2 s_2) = \mathbb{E}\{(D_{\mathrm{F}} - q_1)s_2\} = \mathbb{E}(D_{\mathrm{F}} s_2).$$

Finally, ψ does not depend on π_1 or π_2, and the inner products of D_{seq} with their score components are zero; this is verified in Appendix C. By Lemma 1 the five families of scores exhaust the tangent space, so Equation (10) represents the derivative along every tangent direction and D_{seq} is a gradient. Because it also lies in the tangent space, it is the canonical gradient, and it is the unique influence function of ψ in $\mathcal{M}$; in particular it has minimum variance among regular influence functions (Tsiatis, 2006). □

Remark 1. The proof is a two-stage specialization of general monotone-MAR/CAR theory (Robins et al., 1994; Gill et al., 1997; Barnwell and Chaudhuri, 2025). Its purpose is to establish rigorously that the causal pseudo-outcome in Equation (6), after sequential projection, yields the efficient observed-data gradient for this particular ATE problem.

4 Estimation, robustness, and inference

4.1 Cross-fitted one-step estimator

Let $O_1, \ldots, O_n$ be independent and identically distributed observations from P_O, and let $\eta = (e, \mu_0, \mu_1, \pi_1, \pi_2, Q_0, Q_1)$. With nuisance estimates trained outside observation i's validation fold, define

$$\widehat{H}_i = \widehat{Q}_{0,i} + \frac{R_{1i}}{\widehat{\pi}_{1,i}}(\widehat{Q}_{1,i} - \widehat{Q}_{0,i}) + \frac{R_{2i}}{\widehat{\pi}_{1,i}\widehat{\pi}_{2,i}}(\widehat{G}_i - \widehat{Q}_{1,i}), \tag{11}$$

and

$$\widehat{\psi}_{\text{seq}} = \frac{1}{n}\sum_{i=1}^{n} \widehat{H}_i. \tag{12}$$

An estimated standard error is

$$\widehat{\text{se}}(\widehat{\psi}_{\text{seq}}) = \left[\frac{1}{n(n-1)}\sum_{i=1}^{n}(\widehat{H}_i - \widehat{\psi}_{\text{seq}})^2\right]^{1/2}. \tag{13}$$

One practical training sequence is: fit π_1 to all training records; fit π_2 among $R_1 = 1$ records; fit the causal regressions on complete records with inverse-response weights to reconstruct the full law; regress $\widehat{G}$ on Z_1 among complete records to obtain Q_1; and regress the stage-two augmented pseudo-outcome on Z_0 among $R_1 = 1$ records to obtain Q_0. Other learners are possible, provided they estimate the same nuisance functions and the positivity and convergence conditions below are respected.

4.2 Exact drift and sequential multiple robustness

For arbitrary fixed functions $\bar{G}, \bar{Q}_0, \bar{Q}_1, \bar{\pi}_1, \bar{\pi}_2$ with positive response probabilities and finite displayed expectations, define

$$Q_1^{\bar{G}} = \mathbb{E}(\bar{G} \mid Z_1), \qquad Q_0^{\bar{G}} = \mathbb{E}(\bar{G} \mid Z_0),$$

and let $H(\bar{\eta})$ denote Equation (11) with bars in place of hats.

Proposition 2 (Exact missingness-layer drift). *Let all expectations be taken under the true law P, and let $Q_0^{\bar{G}}$ and $Q_1^{\bar{G}}$ denote true conditional expectations of the working pseudo-outcome $\bar{G}$. Under*

Assumption 2 alone, and with no requirement that any working nuisance be correct,

$$\mathbb{E}\{H(\bar{\eta})\} - \mathbb{E}(\bar{G}) = \mathbb{E}\left[\left(1 - \frac{\pi_1}{\bar{\pi}_1}\right)(\bar{Q}_0 - Q_0^{\bar{G}})\right] + \mathbb{E}\left[\frac{\pi_1}{\bar{\pi}_1}\left(1 - \frac{\pi_2}{\bar{\pi}_2}\right)(\bar{Q}_1 - Q_1^{\bar{G}})\right]. \tag{14}$$

If $\bar{G}$ is formed from $(\bar{e}, \bar{\mu}_0, \bar{\mu}_1)$ as in Equation (6), then

$$\mathbb{E}(\bar{G}) - \psi = \mathbb{E}\left[\frac{\bar{e} - e}{\bar{e}}(\bar{\mu}_1 - \mu_1) + \frac{\bar{e} - e}{1 - \bar{e}}(\bar{\mu}_0 - \mu_0)\right]. \tag{15}$$

Proof. Taking expectation of $H(\bar{\eta})$ over the response indicators conditional on the full data replaces R_1 by π_1 and R_2 by $\pi_1\pi_2$. Subtracting $\mathbb{E}(\bar{G})$ and adding and subtracting $Q_j^{\bar{G}}$ yields Equation (14); the terms involving $Q_j^{\bar{G}}$ reduce to conditional expectations of $\bar{G}$. For Equation (15), condition the barred version of Equation (6) on V, use $\mathbb{E}(A \mid V) = e(V)$ and $\mathbb{E}(Y \mid A = a, V) = \mu_a(V)$, then collect products of the treatment- and outcome-regression errors. □

Corollary 1 (Sequential multiple robustness)**.** *The estimating equation is unbiased for ψ under every nuisance configuration satisfying all three clauses:*

(i) $\bar{e} = e$ *or* $(\bar{\mu}_0, \bar{\mu}_1) = (\mu_0, \mu_1)$;

(ii) $\bar{\pi}_1 = \pi_1$ *or* $\bar{Q}_0 = Q_0^{\bar{G}}$;

(iii) $\bar{\pi}_2 = \pi_2$ *or* $\bar{Q}_1 = Q_1^{\bar{G}}$.

This gives up to $2 \times 2 \times 2 = 8$ sufficient configurations. The result is a layered or sequential form of multiple robustness. The alternatives concern the true projections of the working pseudo-outcome; a particular regression specification is not guaranteed to attain each configuration, and an arbitrary subset of nuisance models cannot be misspecified without restriction.

4.3 Asymptotic normality

Theorem 2 (Cross-fitted asymptotic linearity)**.** *Use a fixed number of folds. Suppose, uniformly across folds: (i) fitted observation probabilities and treatment probabilities are bounded away from zero (and the treatment probability from one); (ii) $\left\|\widehat{H} - H(\eta)\right\|_2 = o_p(1)$ and the influence functions have uniformly integrable $2 + \delta$ moments for some $\delta > 0$; and (iii)*

$$\begin{aligned}
&\|\widehat{e} - e\|_2 \{\|\widehat{\mu}_1 - \mu_1\|_2 + \|\widehat{\mu}_0 - \mu_0\|_2\} = o_p(n^{-1/2}), \\
&\|\widehat{\pi}_1 - \pi_1\|_2 \left\|\widehat{Q}_0 - Q_0^{\widehat{G}}\right\|_2 = o_p(n^{-1/2}), \\
&\|\widehat{\pi}_2 - \pi_2\|_2 \left\|\widehat{Q}_1 - Q_1^{\widehat{G}}\right\|_2 = o_p(n^{-1/2}).
\end{aligned}$$

Then

$$\sqrt{n}(\widehat{\psi}_{\mathrm{seq}} - \psi) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} D_{\mathrm{seq}}(O_i) + o_p(1) \rightsquigarrow N\{0, \mathrm{Var}(D_{\mathrm{seq}})\}.$$

The variance estimator in Equation (13) is consistent, and the estimator attains the semiparametric efficiency bound.

Proof. Conditional on the training samples, cross-fitting makes each validation-fold nuisance fit fixed. Decompose $\widehat{\psi} - \psi$ into the empirical mean of the true influence function, an empirical-process difference on the validation fold, and the conditional drift. The L_2 convergence in (ii) makes the second term $o_p(n^{-1/2})$ without Donsker conditions. Applying Proposition 2 fold by fold and Cauchy–Schwarz bounds the drift by the three products in (iii), up to constants from positivity. Thus the remainder is $o_p(n^{-1/2})$. The central limit theorem applies to the first term. Consistency of the empirical second moment follows from (ii) and uniform integrability. Efficiency follows from Theorem 1. This is the standard cross-fitting argument (Chernozhukov et al., 2018) specialized to Equation (8). □

The conditions in (iii) constrain products of errors, not individual rates, so one factor may converge slowly if its partner converges quickly. The familiar $n^{-1/4}$ heuristic needs care: two factors that are exactly $O_p(n^{-1/4})$ give a product that is $O_p(n^{-1/2})$, which does *not* satisfy (iii). A correct simple sufficient condition is that each factor be $o_p(n^{-1/4})$, or more generally that the two rates r_1, r_2 satisfy $r_1 r_2 = o(n^{-1/2})$. Very small response or treatment probabilities remain a practical threat even when formal positivity holds, motivating weight diagnostics, truncation sensitivity analyses, and estimators targeted for finite-sample stability.

5 Exact cost of complete-case coarsening

Let $S = R_2$ and suppose the analyst replaces (R_1, R_2) by the single complete-case indicator S, so that the working data are $O^{\mathrm{cc}} = (Z_0, S, SL_1, SL_2)$. Such an analyst cannot use π_2, which is a function of L_1 and so is unavailable exactly on the records that S discards; the estimable response functional is instead

$$\rho(Z_0) = \mathbb{P}(S = 1 \mid Z_0) = \pi_1(Z_0)\, \mathbb{E}\{\pi_2(Z_1) \mid Z_0\}. \tag{16}$$

The next restriction creates a submodel in which both approaches identify the same target and, in addition, $\rho = \pi_1 \pi_2$.

Assumption 3 (Common-validity submodel)**.** *The second-stage response probability does not depend on the intermediate confounder after Z_0:*

$$\pi_2(Z_1) = \pi_2(Z_0).$$

Then $S \perp\!\!\!\perp (L_1, L_2) \mid Z_0$ and $\rho(Z_0) = \pi_1(Z_0)\pi_2(Z_0)$, so CCMAR holds. The efficient influence function based only on O^{cc} is

$$D_{\mathrm{cc}}(O^{\mathrm{cc}}) = Q_0 - \psi + \frac{S}{\pi_1\pi_2}(G - Q_0). \tag{17}$$

Theorem 3 (Efficiency loss from collapsing the pattern). *Under Assumptions 1 to 3, with $D_{\mathrm{F}} \in L_2(P)$,*

$$D_{\mathrm{cc}} - D_{\mathrm{seq}} = \frac{R_1}{\pi_1}\left(\frac{C_2}{\pi_2} - 1\right)(Q_1 - Q_0), \tag{18}$$

and this difference is orthogonal to D_{seq}. Consequently,

$$\boxed{\mathrm{Var}(D_{\mathrm{cc}}) - \mathrm{Var}(D_{\mathrm{seq}}) = \mathbb{E}\left[\frac{1-\pi_2(Z_0)}{\pi_1(Z_0)\pi_2(Z_0)}\{Q_1(Z_1) - Q_0(Z_0)\}^2\right] \geq 0.} \tag{19}$$

Proof. Expanding Equation (17) and subtracting Equation (8) gives Equation (18). Let $\Delta = Q_1 - Q_0$ and $\varepsilon = G - Q_1$. Conditional on Z_1, $\mathbb{E}(\varepsilon \mid Z_1) = 0$; conditional on $Z_1, R_1 = 1$, $\mathbb{E}(C_2/\pi_2 - 1) = 0$. Multiplying Equation (18) by the three orthogonal increments in Equation (8) and applying these conditional-mean properties shows $\mathbb{E}\{(D_{\mathrm{cc}} - D_{\mathrm{seq}})D_{\mathrm{seq}}\} = 0$. Finally,

$$\mathbb{E}\{(D_{\mathrm{cc}} - D_{\mathrm{seq}})^2 \mid Z_1\} = \frac{1-\pi_2(Z_0)}{\pi_1(Z_0)\pi_2(Z_0)}\Delta^2,$$

because R_1 and C_2 are Bernoulli with the stated sequential probabilities. Pythagoras gives Equation (19). □

Corollary 2 (When the gain is strict). *The pattern-aware efficiency bound is strictly smaller if and only if*

$$\mathbb{P}\{\pi_2(Z_0) < 1 \text{ and } Q_1(Z_1) \neq Q_0(Z_0)\} > 0.$$

Equivalently, strict improvement requires second-stage missingness and residual predictive information in L_1 about the full-data influence function after conditioning on Z_0.

When π_2 depends on L_1, the complete-case probability conditional on Z_0 averages over an unobserved determinant of selection, and $S \perp\!\!\!\perp (L_1, L_2) \mid Z_0$ generally fails. Then Equation (19) is not an efficiency comparison: the coarsened estimator may target the wrong observed-data functional. This is the distinction between an *information loss* under common validity and an *identification failure* under sequential MAR alone.

Outside Assumption 3 the coarsened analyst must also replace the two quantities in Equation (17) that are no longer estimable from O^{cc}: $\pi_1\pi_2$ becomes $\rho(Z_0)$ of Equation (16), and $Q_0(Z_0) = \mathbb{E}(G \mid Z_0)$ becomes the complete-case projection

$$Q_0^{\mathrm{cc}}(Z_0) = \mathbb{E}(G \mid Z_0, S = 1) = \frac{\mathbb{E}\{\pi_2(Z_1)Q_1(Z_1) \mid Z_0\}}{\mathbb{E}\{\pi_2(Z_1) \mid Z_0\}},$$

which coincides with Q_0 under Assumption 3 and differs from it otherwise. Neither substitution generally restores consistency. With the true causal pseudo-outcome and ρ, either choice of augmentation has population mean $\mathbb{E}(Q_0^{\mathrm{cc}})$: conditional expectation given Z_0 cancels the augmentation. Section 6 reports both this bias calculation and oracle simulations. The estimated-nuisance coarsened procedure also fits its causal regressions using complete records weighted by $1/\widehat{\rho}$; outside common validity, those regressions need not estimate the full-law causal nuisances.

6 Simulation studies

6.1 Gaussian stress design and response mechanisms

The first design isolates the missingness comparison with oracle nuisance functions, then examines the effect of estimating those functions. There are no fully observed baseline covariates W. The full-data law is specified retrospectively:

$$\begin{aligned}
A &\sim \mathrm{Bernoulli}(0.5),\\
Y \mid A = 0 &\sim \mathrm{Beta}(2,4), \qquad Y \mid A = 1 \sim \mathrm{Beta}(4,2),\\
L_1 \mid A, Y &\sim \mathrm{Bernoulli}\{\mathrm{expit}(-0.6 + 0.5A + 0.25Y + 0.1AY)\},\\
L_2 \mid A, Y, L_1 &\sim N(A + Y + 2.5L_1Y, 1.25^2).
\end{aligned}$$

Refactorization of this law yields $e(V)$ and $\mu_a(V)$ by numerical quadrature. The ATE is $\psi = 0.2558505161$ to ten decimal places; all summaries use the unrounded computed value. Stage-one response is

$$\pi_1 = \mathrm{expit}(2 + 0.1A + 0.1Y).$$

The two stage-two mechanisms are

$$\pi_2 = \begin{cases} \mathrm{expit}(-1.1 + 0.1A + 0.1Y), & \text{common validity,}\\ \mathrm{expit}(-1.9 + 0.1A + 0.1Y + 2.2L_1), & \text{sequential MAR only.}\end{cases}$$

The first permits an efficiency comparison and the second illustrates identification failure after coarsening. The oracle experiment uses 2,000 samples of size 2,500 per mechanism. All three estimators are means of their influence-function pseudo-outcomes, with normal intervals based on the empirical variance within each sample. The coarsened oracle procedure uses the true causal functions and Q_0 but weights by $1/\rho(Z_0)$; the supplement also reports the alternative augmentation Q_0^{cc}. Neither oracle version is a feasible causal estimator outside common validity.

This Gaussian design does not satisfy the uniform treatment-positivity condition in Assumption 1. The unbounded support of L_2 permits treatment probabilities arbitrarily close to zero or one. The weaker moment conditions $\mathbb{E}(1/e) = 4.04$ and $\mathbb{E}\{1/(1-e)\} = 5.65$ nevertheless give a finite full-data efficiency bound because Y is bounded. Deterministic quadrature gives $\mathrm{Var}(G) = 0.27924$ and $\mathbb{E}(G^4) = 105.28$. These features make this a stress design for learning the causal nuisance functions;

it is not an empirical verification of Theorem 2.

6.2 Oracle comparison

Table 2: Oracle simulation in the Gaussian stress design ($n = 2{,}500$, 2,000 replicates per scenario).

Scenario	Estimator	Bias	SD	RMSE	Coverage
Common validity	Full data	-0.00005	0.01043	0.01043	94.95%
	Coarsened	-0.00004	0.01775	0.01775	95.10%
	Sequential	-0.00003	0.01723	0.01722	95.55%
Sequential MAR only	Full data	0.00013	0.01052	0.01051	95.05%
	Coarsened	-0.03301	0.01848	0.03783	45.70%
	Sequential	-0.00013	0.01626	0.01626	94.65%

The target ATE is 0.2558505161. The coarsened column uses true G, Q_0, and ρ. Bias and coverage Monte Carlo standard errors are recorded in the supplement; coverage MCSE is at most 1.12 percentage points.

The centered empirical variance reduction under common validity was 5.83%, with paired-bootstrap Monte Carlo 95% interval 4.03%–7.57% (10,000 resamples). This statistic uses deviations from the replicate means, rather than squared deviations from the true ATE, which would estimate a mean-squared-error reduction.

Deterministic product quadrature gives influence-function variances 0.79524 and 0.76224 for the coarsened and sequential procedures, respectively. Their difference is 0.03300, corresponding to a 4.15% reduction. Doubling the quadrature resolutions changes each variance by less than 10^{-8}, and the numerical residual in Equation (19) is below 10^{-12}. These are converged numerical evaluations, not symbolic integrations. The 4.15% value lies within the simulation interval; the interval's width also shows why a finite Monte Carlo variance ratio should not be equated with the efficiency bound.

In the sequential-only scenario, the coarsened oracle bias was -0.03301 with 45.70% coverage, whereas the sequential bias was -0.00013 with 94.65% coverage. Quadrature gives population coarsened bias -0.03282. Replacing Q_0 by Q_0^{cc} gives simulated bias -0.03302; the two population biases coincide, as shown above. This is an identification comparison.

The population pattern proportions obtained by quadrature are 23.98% complete, 65.09% L_1 only, and 10.93% neither under common validity. The corresponding sequential-only proportions are 31.18%, 57.89%, and 10.93%.

6.3 Sensitivity to the second-stage response mechanism

The sensitivity experiment keeps the Gaussian full-data law and uses $\pi_2 = \operatorname{expit}(b + 0.1A + 0.1Y + \gamma L_1)$. Under common validity, $\gamma = 0$ and the intercept b is calibrated to population complete-case proportions 0.5117, 0.2400, and 0.1161. In the identification experiment, γ is 0.75, 1.50, or

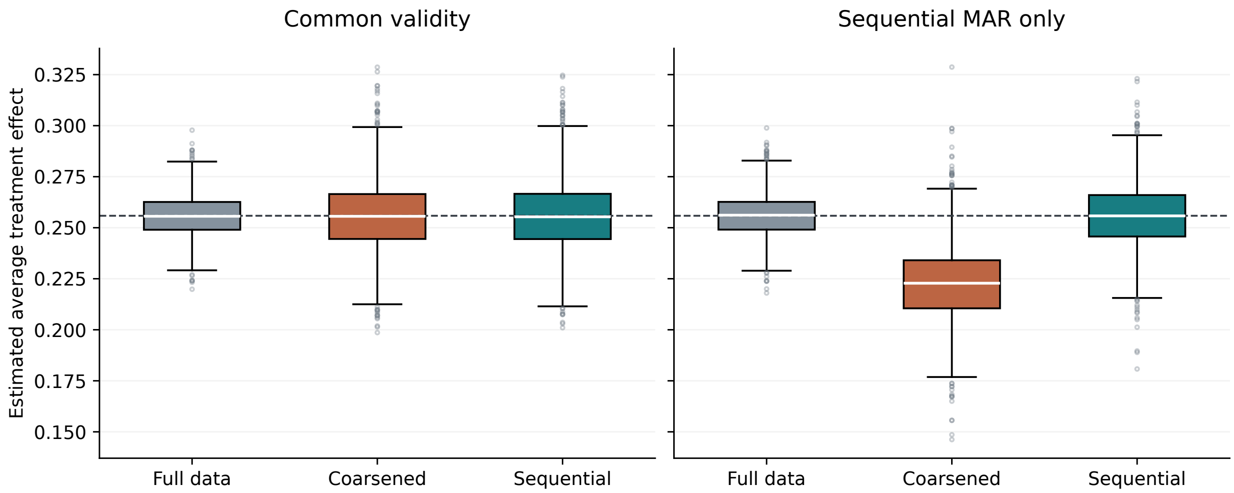

Figure 1: Oracle estimates from the same replicates used in Table 2. Boxes show quartiles, whiskers extend to 1.5 times the interquartile range, and points beyond the whiskers are shown. Dashed lines mark the true ATE.

2.25 and b is calibrated separately to maintain a population complete-case proportion of 0.24. Each configuration uses 5,000 samples of size 2,500. The intercepts, empirical response shares, biases, coverage, and uncertainty in the variance reduction are reported below. Calibration uses deterministic integration; reported empirical shares need not equal the population targets exactly.

Table 3: Efficiency sensitivity under common validity (5,000 replicates per row).

Missingness	Intercept	Complete	L_1 only	Reduction	MC interval
Low	0.19998	51.16%	37.92%	1.47%	0.68–2.25
Moderate	-1.09906	24.01%	65.06%	3.66%	2.52–4.85
High	-2.00020	11.61%	77.45%	5.05%	3.81–6.29

Reduction is $100\{1-\widehat{\mathrm{Var}}(\widehat{\psi}_{\mathrm{seq}})/\widehat{\mathrm{Var}}(\widehat{\psi}_{\mathrm{cc}})\}$, using centered empirical variances. The interval is a paired-bootstrap 95% Monte Carlo interval from 10,000 resamples; it is expressed in percentage points.

Table 4: Identification sensitivity with population complete-case proportion fixed at 24% (5,000 replicates per row).

γ	Intercept	CC bias	CC coverage	Seq. bias	Seq. coverage
0.75	−1.47346	−0.01363	83.22%	0.00011	95.06%
1.50	−1.91220	−0.02633	60.88%	−0.00000	94.90%
2.25	−2.41753	−0.03696	44.28%	−0.00018	95.56%

CC denotes the coarsened oracle procedure using G, Q_0, ρ; Seq. denotes sequential augmentation. Intercepts solve the stated population response-share equation by deterministic integration.

Figure 2: Sensitivity summaries from the same results as Tables 3 and 4. Left: centered empirical variance reduction and paired-bootstrap Monte Carlo intervals. Right: bias and Monte Carlo intervals computed as the estimated bias plus or minus 1.96 Monte Carlo standard errors.

6.4 A design with bounded treatment probabilities

To assess estimated-nuisance behavior under uniform positivity, we also specify a prospective full-data law. Independently, $L_1 \sim \mathrm{Bernoulli}(0.5)$ and $L_2 \sim \mathrm{Uniform}(-1, 1)$. Let

$$\begin{aligned} A \mid L_1, L_2 &\sim \mathrm{Bernoulli}\{\mathrm{expit}(-0.3 + 0.6L_1 + 0.7L_2)\}, \\ \mu_a(L_1, L_2) &= 0.25 + 0.15a + 0.10L_1 + 0.05L_2, \\ Y \mid A, L_1, L_2 &\sim \mathrm{Beta}\{8\mu_A, 8(1 - \mu_A)\}. \end{aligned}$$

Potential outcomes may be generated from these conditional beta laws independently of treatment given the confounders, ensuring conditional exchangeability. The ATE is exactly 0.15, and $\mathrm{expit}(-1) \le e(V) \le \mathrm{expit}(1)$, approximately $[0.269, 0.731]$. Outcomes and response probabilities are also bounded as required for the moment and positivity conditions. The two response mechanisms are the same functions of A, Y, L_1 as above. This design supplies an instance of the identifying assumptions and uniform positivity; it does not by itself verify the nuisance-rate conditions.

All nuisance functions are estimated with the fixed five-fold procedure in Appendix D. Each configuration uses 1,000 replicates. No tuning parameter is selected from the Monte Carlo performance. The full-data benchmark uses all simulated confounders; each incomplete-data estimator receives only its observed covariates.

Table 5: Estimated-nuisance performance with bounded treatment probabilities.

n	Estimator	Bias	SD	Mean SE	RMSE	Coverage
Common validity						
2,500	Full data	-0.00003	0.00661	0.00665	0.00661	95.1%
	Coarsened	-0.00007	0.00935	0.00958	0.00935	95.9%
	Sequential	-0.00001	0.00890	0.00896	0.00890	96.0%
5,000	Full data	-0.00013	0.00455	0.00467	0.00455	95.6%
	Coarsened	-0.00019	0.00607	0.00618	0.00607	95.4%
	Sequential	-0.00027	0.00548	0.00567	0.00548	96.4%
Sequential MAR only						
2,500	Full data	0.00019	0.00675	0.00665	0.00675	95.1%
	Coarsened	0.00437	0.00870	0.00850	0.00973	91.4%
	Sequential	0.00000	0.01227	0.01104	0.01227	95.4%
5,000	Full data	-0.00001	0.00476	0.00467	0.00475	94.0%
	Coarsened	0.00403	0.00574	0.00566	0.00701	87.5%
	Sequential	-0.00003	0.00643	0.00632	0.00643	95.6%

SD is the empirical standard deviation; mean SE is the average estimated standard error. Coverage uses nominal 95% normal intervals. Each configuration uses 1,000 replicates; the true ATE is 0.15. Coverage Monte Carlo SE is at most 1.58 percentage points (about 0.69 near 95% coverage).

Across the four bounded-design configurations, sequential coverage ranges from 95.4% to 96.4%. The bias, empirical variability, and average estimated standard errors are shown together so that coverage can be assessed against its Monte Carlo uncertainty. The common-validity and sequential-only scenarios must still be distinguished when interpreting the coarsened results. A smaller bias does not guarantee a smaller finite-sample mean squared error: in the sequential-only configuration at $n = 2{,}500$, sequential RMSE is 0.01227, compared with 0.00973 for the biased coarsened estimator. The efficiency-bound comparison does not establish a uniform finite-sample MSE ordering.

6.5 Estimated nuisances in the stress design

The corresponding Gaussian-design results appear in Appendix D, including sample sizes 2,500 through 20,000 and oracle substitutions for selected nuisance layers. With all nuisances estimated, sequential coverage ranges from 83.8% to 89.6% across the eight sample-size and scenario configurations. The observed undercoverage is a limitation of this estimator and learner in that design; a reduction in bias or improvement with sample size does not establish the asymptotic rate conditions. The oracle-layer comparisons are diagnostics of the specified fitting procedure, rather than a general decomposition of all sources of finite-sample error.

7 Discussion

The exact variance identity gives a direct interpretation of partial-record value. The loss from coarsening depends on the residual information $(Q_1 - Q_0)^2$ and the factor $(1 - \pi_2)/(\pi_1 \pi_2)$. Holding the full-data law and π_1 fixed, decreasing π_2 increases the absolute variance gap wherever $Q_1 \neq Q_0$. Weak response positivity can nevertheless increase both efficiency bounds and make finite-sample estimation unstable. Many L_1-only records yield no gain when $Q_1 = Q_0$; when $\pi_2 = 1$, the gap is zero regardless of the predictive value of L_1.

The model comparison also clarifies assumption strength. Under Assumption 3, complete-case coarsening wastes information but remains valid. If L_1 affects the chance that L_2 is observed, sequential MAR may remain plausible because that chance depends on information observed at stage two, whereas CCMAR generally fails after L_1 is hidden inside the all-complete indicator. In that setting, the pattern-aware estimator is not simply a more precise version of the CCMAR estimator; it is based on a different identifying model.

The present results sit between two literatures. General sequential augmentation and efficiency under monotone attrition are well established (Robins et al., 1994; Gill et al., 1997; Barnwell and Chaudhuri, 2025). Recent causal work addresses complete-case MAR (Levis et al., 2025), multiple missing confounders or exposures under other MAR/MNAR restrictions (Wen and McGee, 2026), and broad numerical comparisons of practical missing-confounder estimators (Benz et al., 2025; Williamson et al., 2026). The narrow value added here is the exact causal score comparison between retaining and collapsing a monotone intermediate pattern. We do not solve the nonmonotone 2^q-pattern projection problem under the original CCMAR model, and we do not claim the generic monotone-MAR influence-function transformation as new.

The ordered-confounder construction in Section 4.2 of Wen and McGee (2026) already uses sequential observation probabilities and iterated projections to obtain an influence function and a targeted maximum likelihood estimator. That construction allows an incompletely observed exposure and imposes outcome-independent missingness restrictions. Here the exposure is fully observed, while both response hazards may depend on the observed treatment and outcome. Our comparison quantifies the efficiency loss from collapsing the intermediate pattern on the common-validity submodel and distinguishes that loss from bias when coarsening invalidates identification. The different identifying restrictions are essential to this comparison; an ordering of confounders alone does not distinguish the present work from the existing construction.

The two estimated-nuisance designs serve different purposes. The bounded-propensity design satisfies uniform positivity and assesses the implementation in a regular setting. The Gaussian design has a finite oracle efficiency bound but violates uniform treatment positivity; it exposes substantial sensitivity to nuisance fitting and poor interval calibration. Neither experiment establishes the product-rate conditions in Theorem 2. In particular, fitting a fixed spline basis does not by itself establish the required nuisance convergence rates. All results are synthetic, and the simulation comparison uses one specified learner and two missingness mechanisms. It does not establish performance across arbitrary nuisance misspecification or relative to multiple imputation and other practical alternatives.

The variance difference in Equation (19) is a first-order efficiency-bound comparison. Finite-sample nuisance error can obscure or reverse the corresponding ordering of estimator variances. The normal intervals in the stress design should therefore not be treated as reliable at the sample sizes studied. A bootstrap or a nuisance-aware variance procedure would require its own validation; the present results do not establish that either would repair the undercoverage.

Several extensions would strengthen the practical case: a real-data or plasmode application with defensible acquisition ordering; broader learner and nuisance-misspecification comparisons; and a validated inference procedure for difficult overlap settings. Monotonicity rules out L_2-only records and may not suit unstructured electronic health records. Neither sequential MAR nor causal exchangeability is testable from the observed data alone, so the identifying model must be justified by substantive knowledge of data collection. Extension to more monotone stages is also of interest, but is not established here.

Data and code availability

All data analyzed in this manuscript are synthetic. The simulation programs, configuration and environment records, complete replicate-level results, deterministic variance check, figure and table generators, and editable manuscript source are publicly available at `https://github.com/KeivanBolouri/pattern-aware-sequential-mar`. The repository also includes example synthetic datasets and a generator that recreates the full and observed covariates and fold assignments for any estimated-nuisance replicate using its recorded seed. The README documents how to verify the archived results, regenerate the simulations, and rebuild the manuscript. The original CCMAR software of Levis et al. (2025) is cited as prior software at `https://github.com/alexlevis/flex-ate-confounders-MAR`; the sequential code is separate.

Reproducibility

Every numerical table and figure is generated from the result files in the supplement. Simulation streams are indexed by configuration and replicate; the base seed for estimated nuisances is 20260910. The README specifies all streams and software versions. The implementation accepts missing covariates as unavailable values and evaluates each augmentation only on the required observation pattern. Regression checks verify invariance to unavailable covariates, oracle response probabilities in training weights, equality with the directly evaluated oracle estimating equation, and centered variance calculations. Numerical quadrature is checked at two resolutions. These checks address specific implementation risks; they do not prove finite-sample coverage or the nuisance-rate assumptions of the asymptotic theorem.

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A Tangent space of the observed-data model

Theorem 1 verifies that D_{seq} represents the pathwise derivative of ψ along five families of score directions. That verification is exhaustive only if those families span the tangent space of $\mathcal{M}$. The following lemma supplies the missing step; it is the two-stage case of the coarsening-at-random tangent-space decomposition (Robins et al., 1994; Gill et al., 1997; Tsiatis, 2006), written out here because the causal argument depends on it.

Lemma 1 (Orthogonal decomposition of $L_2^0(P_O)$). *Let Assumption 2 hold and suppose in addition $\epsilon_\pi \le \pi_1(Z_0) \le 1-\epsilon_\pi$ and $\epsilon_\pi \le \pi_2(Z_1) \le 1-\epsilon_\pi$ almost surely, so that all three response patterns occur with positive probability. Then the subspaces $\mathcal{T}_0, \mathcal{T}_{\pi_1}, \mathcal{T}_1, \mathcal{T}_{\pi_2}, \mathcal{T}_2$ defined in the proof of Theorem 1 are closed, mutually orthogonal, and*

$$L_2^0(P_O) = \mathcal{T}_0 \oplus \mathcal{T}_{\pi_1} \oplus \mathcal{T}_1 \oplus \mathcal{T}_{\pi_2} \oplus \mathcal{T}_2.$$

Consequently the tangent space of $\mathcal{M}$ at P_O is all of $L_2^0(P_O)$, and a pathwise differentiable parameter has a unique gradient.

Proof. Orthogonality. All five statements follow from iterated conditioning together with Equations (2) and (3). Two are representative. For $s_0 \in \mathcal{T}_0$ and $R_1 s_1 \in \mathcal{T}_1$,

$$\mathbb{E}(s_0 R_1 s_1) = \mathbb{E}\{s_0\, \mathbb{E}(R_1 s_1 \mid Z_0)\} = \mathbb{E}\{s_0 \pi_1 \mathbb{E}(s_1 \mid Z_0)\} = 0,$$

where $\mathbb{E}(s_1 \mid Z_0, R_1 = 1) = \mathbb{E}(s_1 \mid Z_0)$ by Equation (2). For $R_1 s_1 \in \mathcal{T}_1$ and $R_2 s_2 \in \mathcal{T}_2$, using $R_1 R_2 = R_2$ and conditioning first on Z_1,

$$\mathbb{E}(R_1 s_1 R_2 s_2) = \mathbb{E}\{s_1\, \mathbb{E}(R_2 s_2 \mid Z_1)\} = \mathbb{E}\{s_1 \pi_1 \pi_2 \mathbb{E}(s_2 \mid Z_1)\} = 0,$$

because Equation (3) gives $\mathbb{E}(C_2 \mid Z_1, R_1 = 1, L_2) = \pi_2(Z_1)$ and Equation (2) gives $L_2 \mid (Z_1, R_1 = 1) \sim L_2 \mid Z_1$. The remaining pairs are obtained the same way, using $\mathbb{E}(R_1 - \pi_1 \mid Z_0) = 0$ and $\mathbb{E}\{R_1(C_2 - \pi_2) \mid Z_1\} = 0$.

Completeness. The observed datum is determined by Z_0 on $\{R_1 = 0\}$, by Z_1 on $\{R_1 = 1, C_2 = 0\}$, and by Z_2 on $\{R_2 = 1\}$. Hence any $f \in L_2^0(P_O)$ can be written as

$$f = \alpha(Z_0)\mathbf{1}\{R_1 = 0\} + \beta(Z_1)\mathbf{1}\{R_1 = 1, C_2 = 0\} + \gamma(Z_2)\mathbf{1}\{R_2 = 1\},$$

and the two-sided bounds make α, β, γ square integrable. Define

$$\begin{aligned} h_2 &= \mathbb{E}(\gamma \mid Z_1) - \beta, & s_2 &= \gamma - \mathbb{E}(\gamma \mid Z_1), \\ h_1 &= \mathbb{E}[(1 - \pi_2)\beta + \pi_2\mathbb{E}(\gamma \mid Z_1) \mid Z_0] - \alpha, & s_1 &= \beta + \pi_2 h_2 - \alpha - h_1, \\ s_0 &= \alpha + \pi_1 h_1. \end{aligned}$$

By construction $\mathbb{E}(s_2 \mid Z_1) = 0$ and $\mathbb{E}(s_1 \mid Z_0) = 0$. Writing $D = s_0 + (R_1 - \pi_1)h_1 + R_1 s_1 + R_1(C_2 - \pi_2)h_2 + R_2 s_2$ and evaluating on each pattern gives $D = \alpha$ when $R_1 = 0$; $D = \alpha + h_1 + s_1 - \pi_2 h_2 = \beta$ when $R_1 = 1, C_2 = 0$; and $D = \beta + h_2 + s_2 = \gamma$ when $R_2 = 1$. Thus $D = f$. Finally

$$\mathbb{E}(s_0) = \mathbb{E}\{(1 - \pi_1)\alpha\} + \mathbb{E}\{\pi_1(1 - \pi_2)\beta\} + \mathbb{E}(\pi_1\pi_2\gamma) = \mathbb{E}(f) = 0,$$

the three terms being exactly the pattern probabilities implied by Assumption 2. Uniqueness of the representation follows from orthogonality. □

Remark 2 (Boundary cases)**.** Only the completeness half uses the upper bounds on π_1, π_2, and their role is solely to guarantee that each of the three response patterns occurs with positive probability, so that α, β, γ are identified and square integrable.

The two boundaries behave differently, and it is worth being precise about which subspace degenerates. Suppose $\pi_2 \equiv 1$, so that L_2 is observed whenever L_1 is and $R_2 = R_1$. Then $C_2 - \pi_2 \equiv 0$, so $\mathcal{T}_{\pi_2} = \{0\}$: there is no second response mechanism to score. But $\mathcal{T}_2 = \{R_2 s_2(Z_2) : \mathbb{E}(s_2 \mid Z_1) = 0\}$ is *not* affected, because the conditional law of L_2 given Z_1 is still a free component of the model and is still observed on $\{R_1 = 1\}$. The pattern $\{R_1 = 1, C_2 = 0\}$ is empty, β drops out, and the construction in the proof goes through with $h_2 = 0$, $s_2 = \gamma - \mathbb{E}(\gamma \mid Z_1)$, $s_1 = \mathbb{E}(\gamma \mid Z_1) - \mathbb{E}(\gamma \mid Z_0)$ and $h_1 = \mathbb{E}(\gamma \mid Z_0) - \alpha$, giving $L_2^0(P_O) = \mathcal{T}_0 \oplus \mathcal{T}_{\pi_1} \oplus \mathcal{T}_1 \oplus \mathcal{T}_2$. Symmetrically, $\pi_1 \equiv 1$ removes $\mathcal{T}_{\pi_1}$ and leaves $\mathcal{T}_1$ intact. The influence function in Theorem 1 remains valid. When $\pi_2 \equiv 1$, its two inverse-weighted increments combine:

$$\frac{R_1}{\pi_1}(Q_1 - Q_0) + \frac{R_1}{\pi_1}(G - Q_1) = \frac{R_1}{\pi_1}(G - Q_0).$$

The influence function is therefore $Q_0 - \psi + R_1(G - Q_0)/\pi_1$; the residual increment does not vanish. When $\pi_1 \equiv 1$, the first two terms instead combine to $Q_1 - \psi$, leaving $Q_1 - \psi + R_2(G - Q_1)/\pi_2$.

When $\pi_2 = 1$ on a set of positive probability rather than everywhere, the same argument applies on that set and the general one on its complement. There Equation (19) contributes nothing, since its integrand carries the factor $1 - \pi_2$; this is the degenerate boundary of Corollary 2.

B Additional derivation of the efficiency identity

This appendix makes explicit the orthogonality used in Theorem 3. Let

$$\Delta = Q_1 - Q_0, \qquad \varepsilon = G - Q_1,$$

so $\mathbb{E}(\Delta \mid Z_0) = 0$ and $\mathbb{E}(\varepsilon \mid Z_1) = 0$. Under Assumption 3,

$$\begin{aligned} D_{\text{seq}} &= Q_0 - \psi + \frac{R_1}{\pi_1}\Delta + \frac{R_1 C_2}{\pi_1 \pi_2}\varepsilon, \\ U := D_{\text{cc}} - D_{\text{seq}} &= \frac{R_1}{\pi_1}\left(\frac{C_2}{\pi_2} - 1\right)\Delta. \end{aligned}$$

The product of U with $Q_0 - \psi$ has mean zero by conditioning on Z_1, R_1 and averaging over C_2. The product with $R_1\Delta/\pi_1$ also has mean zero for the same reason. For the last term,

$$\mathbb{E}\left[U \frac{R_1 C_2}{\pi_1 \pi_2}\varepsilon\right] = \mathbb{E}\left[\frac{1-\pi_2}{\pi_1\pi_2}\Delta\varepsilon\right] = 0,$$

because the coefficient and Δ are measurable with respect to Z_1, whereas $\mathbb{E}(\varepsilon \mid Z_1) = 0$. Thus $U \perp D_{\text{seq}}$. Moreover,

$$\begin{aligned} \mathbb{E}(U^2 \mid Z_1) &= \frac{\Delta^2}{\pi_1^2}\mathbb{E}\left[R_1\left(\frac{C_2}{\pi_2} - 1\right)^2 \middle| Z_1\right] \\ &= \frac{1-\pi_2}{\pi_1\pi_2}\Delta^2, \end{aligned}$$

which establishes the identity.

C Observed-score calculations for response mechanisms

For completeness, consider $S_{\pi_1} = (R_1 - \pi_1)h_1(Z_0)$. Using Equation (10), each inner product $\mathbb{E}(D_{\text{seq}} S_{\pi_1})$ is zero: terms without R_1 vanish because $\mathbb{E}(R_1 - \pi_1 \mid Z_0) = 0$, while terms containing the inverse-weighted residual increments vanish after conditioning successively on Z_0 and Z_1. For stage two, write the observed score as

$$S_{\pi_2} = (R_2 - R_1\pi_2)h_2(Z_1) = R_1(C_2 - \pi_2)h_2(Z_1).$$

The q_0 and $q_1 - q_0$ components are orthogonal because the centered Bernoulli residual has conditional mean zero. For the terminal component,

$$\mathbb{E}\left[\frac{R_2}{\pi_1\pi_2}(D_{\text{F}} - q_1)(R_2 - R_1\pi_2)h_2(Z_1)\right]$$

equals, after response averaging,

$$\mathbb{E}[(1-\pi_2(Z_1))(D_{\mathrm{F}}-q_1)h_2(Z_1)] = 0,$$

because $\mathbb{E}(D_{\mathrm{F}}-q_1 \mid Z_1) = 0$. Thus the canonical gradient carries no component in either response-mechanism tangent space, as required because the causal target is a functional of the full-data law alone.

D Cross-fitted estimation with estimated nuisances

Observed-data implementation

Each sample is randomly divided into five folds. All basis transformations and fitted models used on a validation fold are trained on the other four folds. The outcome basis uses all training values of Y. The incomplete-data causal basis uses only L_2 values with $R_2 = 1$ in the training set. A separate full-data benchmark chooses its own basis using all training L_2 values. Missing covariates are represented by unavailable values: the implementation reads L_1 only when $R_1 = 1$ and L_2 only when $R_2 = 1$. There is no fitting, prediction, or knot selection using hidden simulated confounders in the incomplete-data procedures.

Continuous coordinates use cubic B-splines with six empirical quantile knots, including the boundary knots, and no spline bias column. Each fitted regression includes an intercept. The design for Z_0 contains the spline in Y, A, and their interactions. The design for Z_1 is saturated in (A, L_1), including their products with the spline in Y. The causal design contains the spline in L_2, L_1, and their interactions. Logistic regressions use an ℓ_2 penalty with inverse regularization parameter $C = 1$; ridge regressions use penalty 10^{-3}. Hyperparameters are fixed across designs, sample sizes, and replicates. Logistic fitting allows at most 2,000 iterations, and a convergence warning stops the run instead of being silently suppressed.

The response models are fitted on all training records for π_1 and ρ, and on $R_1 = 1$ records for π_2. Causal regressions are fitted on complete records with weights $1/(\widehat{\pi}_1\widehat{\pi}_2)$ for the sequential procedure and $1/\widehat{\rho}$ for the coarsened procedure. The sequential Q_1 regression uses complete records. Its Q_0 regression uses the augmented stage-two pseudo-outcome on training records with $R_1 = 1$. The coarsened projection regression uses complete records. These projection regressions use causal predictions fitted within the same outer training fold; there is no additional inner cross-fitting. The held-out fold is excluded from all these training operations.

Fitted treatment probabilities are truncated to $[0.01, 0.99]$, and fitted response probabilities to $[0.01, 1]$. Oracle substitutions are evaluated without truncation. In the stress design the true propensity can lie outside the fitted interval, so fixed truncation does not preserve the true propensity and cannot be assumed asymptotically harmless. In the bounded design the truncation thresholds lie outside the true propensity range. No truncation-sensitivity result is claimed without a corresponding run in the supplement.

Behavior across sample sizes

Table 6: Estimated-nuisance performance in the Gaussian stress design.

n	Estimator	Bias	SD	Mean SE	RMSE	Coverage
Common validity						
2,500	Full data	0.00023	0.01193	0.00947	0.01193	87.0%
	Coarsened	−0.00076	0.04699	0.03016	0.04697	84.0%
	Sequential	−0.00135	0.04827	0.03087	0.04826	84.0%
5,000	Full data	0.00004	0.00761	0.00665	0.00760	91.2%
	Coarsened	0.00048	0.02590	0.01754	0.02589	85.6%
	Sequential	0.00053	0.02712	0.01786	0.02711	85.7%
10,000	Full data	0.00042	0.00573	0.00471	0.00574	88.0%
	Coarsened	−0.00028	0.01335	0.00968	0.01334	85.0%
	Sequential	−0.00033	0.01325	0.00965	0.01324	84.8%
20,000	Full data	−0.00020	0.00417	0.00337	0.00417	88.8%
	Coarsened	−0.00012	0.00795	0.00628	0.00794	87.6%
	Sequential	−0.00016	0.00792	0.00619	0.00791	88.4%
Sequential MAR only						
2,500	Full data	0.00040	0.01189	0.00954	0.01189	87.8%
	Coarsened	−0.03198	0.03702	0.02398	0.04891	61.9%
	Sequential	0.00104	0.05048	0.03204	0.05047	83.8%
5,000	Full data	0.00027	0.00784	0.00671	0.00784	91.2%
	Coarsened	−0.03238	0.01761	0.01308	0.03686	32.2%
	Sequential	0.00076	0.02027	0.01546	0.02027	87.8%
10,000	Full data	0.00005	0.00536	0.00473	0.00536	92.4%
	Coarsened	−0.03283	0.01057	0.00843	0.03449	8.0%
	Sequential	−0.00006	0.01451	0.01013	0.01450	85.8%
20,000	Full data	0.00000	0.00386	0.00336	0.00385	90.0%
	Coarsened	−0.03300	0.00697	0.00575	0.03372	0.0%
	Sequential	0.00016	0.00827	0.00626	0.00825	89.6%

SD is the empirical standard deviation; mean SE is the average estimated standard error. Coverage uses nominal 95% normal intervals. Replicate counts are 1,000, 1,000, 500, and 250 at increasing sample sizes. Coverage Monte Carlo SE is at most 1.58, 1.58, 2.24, and 3.16 percentage points, respectively.

Under common validity, the ratio of mean estimated SE to empirical SD for the sequential estimator is 0.64, 0.66, 0.73, 0.78 at the four increasing sample sizes. These ratios assess the scale of interval miscalibration directly. The finite-sample precision comparison is given by the empirical SDs, whereas Equation (19) compares first-order efficiency bounds. They need not agree at these sample sizes. Outside common validity, the coarsened estimator's bias reflects its invalid identifying model as well as nuisance fitting.

Table 7: Oracle-substitution diagnostics in the Gaussian design ($n = 2{,}500$).

Scenario	Known	Seq. bias	Seq. SD	Seq. cov.	CC bias	CC SD	CC cov.
Common	None	-0.00135	0.04827	84.0%	-0.00076	0.04699	84.0%
	Causal	0.00044	0.01941	96.0%	0.00077	0.01877	96.6%
	Response	-0.00015	0.04446	82.1%	0.00009	0.04454	82.8%
	All	0.00076	0.01789	96.8%	0.00107	0.01830	97.6%
Sequential	None	0.00104	0.05048	83.8%	-0.03198	0.03702	61.9%
	Causal	-0.00037	0.01934	95.5%	-0.03241	0.01832	49.4%
	Response	0.00193	0.04868	82.2%	-0.03158	0.03625	59.2%
	All	-0.00113	0.01660	95.0%	-0.03321	0.01855	45.6%

Causal: true e, μ_0, μ_1. Response: true π_1, π_2, ρ throughout training and validation. All: true causal, response, and projection functions, with Q_0^{cc} for the coarsened procedure. There are 1,000 replicates per row except 500 for All. Corresponding configurations share random-number streams.

Oracle substitutions and implementation checks

The causal-oracle option substitutes the true e and both outcome regressions wherever G is evaluated, including the training projections. The response-oracle option substitutes true π_1, π_2, ρ in the causal training weights, the training pseudo-outcome for Q_0, and the final validation-fold augmentation. The all-oracle option also replaces the projection functions. True projection functions are projections of the true causal pseudo-outcome, so that option requires the causal-oracle option. Under sequential MAR alone, the coarsened all-oracle diagnostic uses Q_0^{cc}, the complete-case conditional projection.

For the sequential procedure under common validity, the empirical SD is 0.04827 with all nuisances estimated, 0.01941 with the causal functions known, and 0.04446 with the response functions known. These comparisons identify sensitivity to the specified nuisance fitting choices; they do not show that every remaining discrepancy is caused by a single layer. Shared replicate seeds make the comparisons paired, and all replicate-level results are retained.

The supplied regression checks verify invariance to hidden covariate values, oracle training weights, and equality of the all-oracle calculation with Equation (8). They also distinguish variance from MSE. These checks target implementation correctness; an all-oracle match alone neither validates estimated nuisances nor establishes nominal coverage.